

<https://rupakm.github.io/leslie> <https://github.com/rupakm/leslie> <https://rupakm.github.io/leslie/docs>

Verifying ABA with Leslie

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Abstract

Our main objective is the verification of an implementation of Asynchronous Byzantine Agreement. The proof of correctness consist in building a simulation relation between the implementation and an implementation running in an idealised setting. The simulation ensures that the correctness of the idealised implementation implies the correctness of the original one. However, this implementation is complex, thus, building the simulation in one go is intractable. Fortunately, the implementation is designed as the interlocking of multiple sub-protocols (GBCA, WCC, RSD, Gather, AVSS, BRB). This architecture allows us to decompose the construction of the overall simulation into smaller steps. Each implementation of sub-protocols will be simulated by an idealised version. Thus, when building the simulation for a protocol using sub-protocols, the sub-implementations will be replaced by their idealised versions; making the construction easier.